

Hexagonal plate current

W21 (2021) — English translation

An electric current flows through the vertices A, B, C and D of a thin metal regular hexagon. The current strengths $2I$ and $5I$ flow into the vertices A and B , respectively, and the current strength $13I$ flows out of the vertex D . The OY axis is directed vertically upward, and the OX axis is directed vertically to the left.

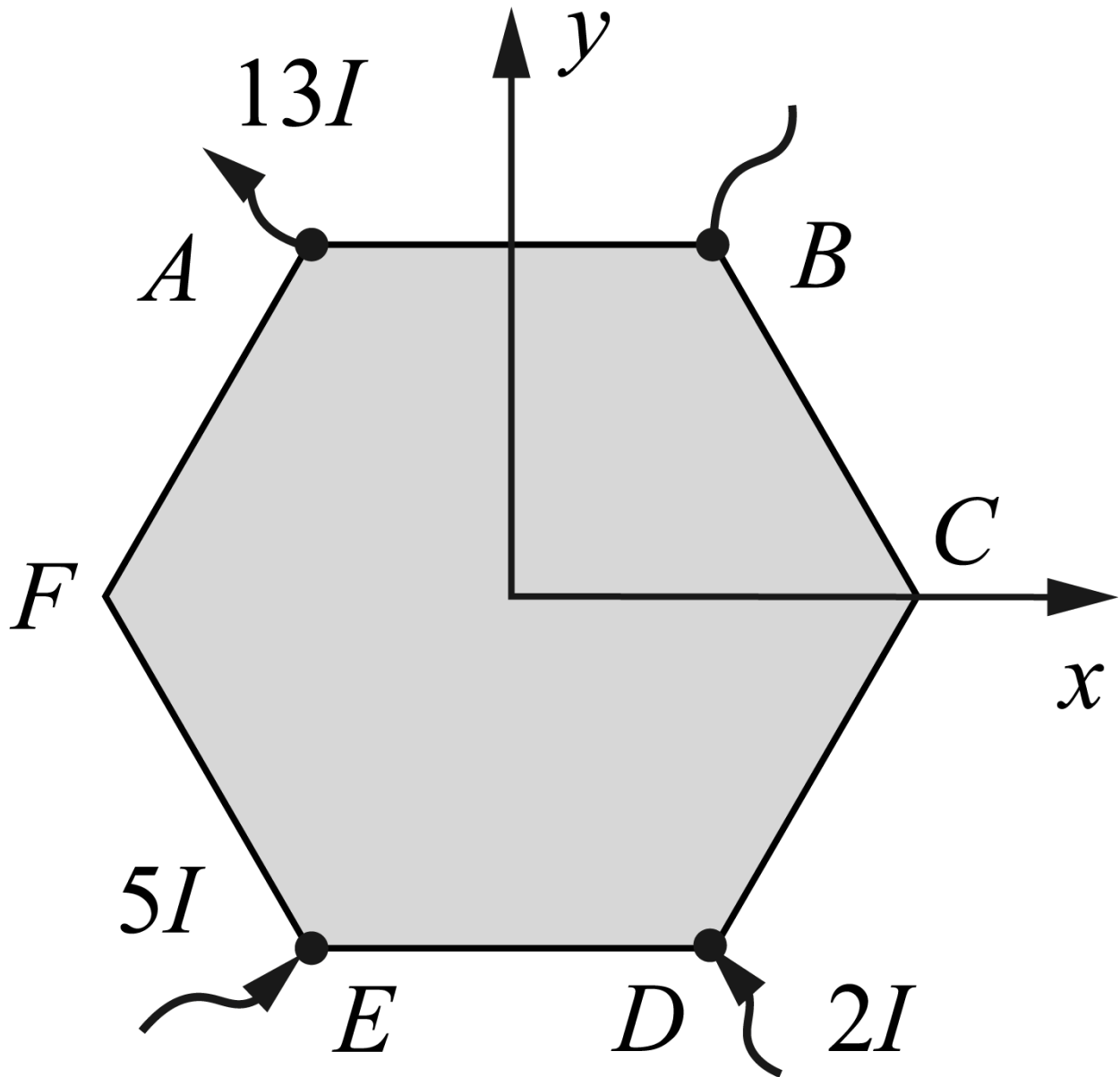


Fig. 1.

A1 Indicate the signs of the electric field projections E_x and E_y at the center of the plate, and **10.00** also find the ratio of their magnitudes.